The six group descriptions in the small chiral atlas

Human Compression project

Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382, Table 1 on p. 376, give four rank-four rows. The previously accepted literal-entry component proves the corresponding eight polytopes, including their duals, and identifies their actual cover groups and rotation quotients. Here we verify the structural names, without a census or an inference from group orders or SmallGroup IDs.

We write C_m for the cyclic group of order m and $E_{27} = C_3^3$. The source's D_9 denotes the dihedral group of order 18. Semidirect products assert the existence of the indicated normal factor and complement; their actions need not be specified uniquely by the names. Every parenthesis below is the parenthesis printed in the source.

Lemma 1 (Finite transport and decomposition)**.** *Let $X = \langle x_1, \dots, x_s \rangle$ and $Y = \langle y_1, \dots, y_s \rangle$ be finite permutation groups. Suppose a map $f : X \to Y$ is defined on every element, sends identity to identity, and satisfies $f(xx_i) = f(x)y_i$ for every x, i. Then f is a surjective homomorphism. If $|X| = |Y|$, it is an isomorphism. If instead its identity fibre is N, it induces $X/N \cong Y$.*

If $N, K \le H$, $N \trianglelefteq H$, $N \cap K = 1$ and $|N||K| = |H|$, then $H \cong N \rtimes K$. If also $[N, K] = 1$, then $H \cong N \times K$.

Proof. Following any word in the generators through the edge identities gives the corresponding word in their images. Every group element has such a word, because inverses are positive powers in a finite group. Appending the word for z to the word for x proves $f(xz) = f(x)f(z)$. The images generate Y. The two conclusions follow from the kernel and order formulas. For the second statement, the normal product NK has order $|N||K|$, hence equals H. Multiplication identifies the corresponding semidirect product with H; commuting factors give the direct product. □

The reader constructs complete generated sets by breadth-first closure. It propagates each proposed map from identity and tests every generator edge, rejecting inconsistent assignments. For a proposed factorization, it checks actual membership, conjugation of each normal-factor generator by every ambient generator, trivial intersection and the exact product order. In the direct-product case it checks commutation of the two generator sets. Thus the lemma applies to the literal subgroups, recursively, rather than merely to their names.

The elementary atoms require no naming database. A cyclic generator is checked by closure. Commuting generators of exponent three and order 27 give E_{27}. The dihedral atom is generated by x, y with $|x| = 9$, $y^2 = 1$, $y^{-1}xy = x^{-1}$ and group order 18. For Q_8, the corresponding tests are $|x| = 4$, $y^2 = x^2$, $y^{-1}xy = x^{-1}$ and group order eight. These relations reduce words to the usual distinct normal forms. Finally an explicit generator-edge isomorphism from the full group of 24 permutations on four letters certifies each S_4 atom.

Theorem 1 (The structural assertions of Table 1)**.** *For the retained rank-four entries, the cover and rotation groups have the following structures. The same conclusions hold for their duals.*

Type	*Cover group*	*Rotation group*
$\{3, 3, 8\}$	$((Q_8 \times C_2) \rtimes C_2) \rtimes S_4$	$Q_8 \rtimes S_4$
$\{3, 6, 9\}$	$E_{27} \rtimes (D_9 \times C_2)$	$(E_{27} \rtimes C_3) \rtimes C_2$
$\{3, 6, 18\}$	$(E_{27} \rtimes (D_9 \times C_2)) \times C_2$	$((E_{27} \rtimes C_3) \rtimes C_2) \times C_2$
$\{6, 6, 9\}$	$(E_{27} \rtimes (D_9 \times C_2)) \times C_2$	$((E_{27} \rtimes C_3) \rtimes C_2) \times C_2$

Proof. For each of the eight accepted entries let $W = \langle r_0, r_1, r_2, r_3 \rangle$ be its literal cover group, $H = \langle r_0 r_1, r_0 r_2, r_0 r_3 \rangle$, and let N be generated by the source words. The checker reconstructs these actual permutations. It verifies a full injective generator-edge map from W to its compact certificate model. It separately propagates the images of the three displayed generators of H and checks that the identity fibre is exactly the literal subgroup N. This identifies the rotation quotient with its compact model, including the actual kernel. It also verifies $[W : H] = 2$ and the stated cover and quotient orders.

The 16 atomic and compound registry models have explicit normal factors, complements and generator-edge transports. Applying the lemma at every registry node proves the six displayed names. All eight actual cover and quotient pairs are transported to these same named models. In particular the last two source rows and their duals use identical registry models; their common names are proved by maps, not by repeated order comparisons. The finite verification has 53 maps and 133,673 generator-edge checks. The actual kernel orders are respectively $2, 3, 3, 3$ for the four source rows. □

Attribution and scope. The examples and structural assertions belong to the source authors. The finite transport/decomposition machinery reuses the accepted atlas method; this application supplies explicit certificates and the exact source joins. GAP proposes witnesses only. The final reader uses finite integer permutations and does not trust GAP's proposed isomorphisms, normal subgroups, complements or SmallGroup identifications. It acquires the earlier proof that these literal W, H, N give the stated polytopes. The original failed output-format attempt is retained separately. No exclusion of other covers, complete chiral census, almost-simple atlas result or whole-source completion is asserted here.